Title: Enhancing Product Quality through Sustainable Manufacturing Practices: A Statistical Quality Control Approach

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1. Abstract:

The presented data consists of measurements from three different peelers (No. 3, No. 5, and No. 6) for specific attributes, including size, wood type, and girth. X-Bar ($\bar{x}$) and R charts have been constructed to monitor the central tendency and variability of the measured characteristics over time.

For each peeler, the data includes girth measurements, squared girth values, and corresponding calculations for X-Bar and R charts. The upper and lower control limits (UCL_x, LCL_x, UCLR, LCLR) have been established to visually assess whether the process is within statistical control. The mean ($\bar{X}$) of the measurements is also calculated.

These control charts serve as valuable tools in quality control, allowing for the identification of any trends, shifts, or outliers in the manufacturing process. The upper and lower control limits help determine if the process is stable, and deviations from these limits may indicate special cause variation that requires further investigation and corrective action.

2. Introduction:

The provided dataset encompasses measurements from three distinct peelers (No. 3, No. 5, and No. 6) in a manufacturing or production setting, focusing on attributes such as size, wood type (poplar), and girth. The objective of this dataset is to employ statistical methods, particularly X-Bar ($\bar{x}$) and R charts, to monitor and assess the variation and central tendency of these measured characteristics over a series of samples.

The X-Bar chart is utilized to track the average or central tendency of the measurements, while the R chart monitors the variability or dispersion within each subgroup. These control charts are fundamental tools in statistical process control (SPC) and are employed to detect any deviations from expected norms in the manufacturing process.

The dataset includes individual measurements, squared values, and the necessary calculations for constructing X-Bar and R charts. Additionally, upper, and lower control limits have been established to aid in the visual interpretation of the data. The mean values are calculated to provide a baseline for comparison.

This introduction sets the stage for the subsequent analysis, emphasizing the importance of employing statistical techniques to ensure the stability and quality of the manufacturing process. The utilization of X-Bar and R charts allows for a systematic assessment of the process, enabling timely identification and corrective action in response to any variations that may impact product quality.

3. Methodology:

3.1 Problem statement:

A plywood manufacturing company, producing plywood for many years. A while ago they've noticed a change in plywood thickness as they've monitored that thickness of newly manufactured plywood is greater than that of standardized ones. They noticed that about 3 Poplar Trees are peeled to make about 15 Plywood. They've checked every department is working fine but the main issue is in peeling department. Now they want to check exactly where from shredding department the error is being done, and whether it's a machine or a human error and how much the number of defective items has been produced this week.

3.2 Data Collection:

Measurements were collected from three different peelers (No. 3, No. 5, and No. 6) for attributes such as girth, size (4 - 3), and wood type (poplar). A Plywood supplier company situated in Haripur Road, Near Hattar Industrial Estate, named Haidery Plywood. It's show room is in 61 – A city Saddar Road, Rawalpindi.

Data Collected (Log Circumference in Inches)

SR. NO.	PEELER NO : 3	PEELER NO : 5	PEELER NO : 6
	SIZE : 4 - 3	SIZE : 4 - 3	SIZE : 4 - 3
	WOOD : POPLAR	WOOD : POPLAR	WOOD : POPLAR
	GIRTH	GIRTH	GIRTH
1	26	20	20
2	22	21	20
3	20	16	24
4	24	18	23
5	19	25	20
6	21	20	20
7	23	21	18
8	20	21	16
9	15	20	19
10	17	20	24

3.3 Construction of X-Bar ($\bar{x}$) and R Charts:

X-Bar charts were created to monitor the average or central tendency of the measurements over time for each peeler. R charts were constructed to monitor the variability or dispersion within each subgroup of measurements. We use Excel software for calculation.

SR.NO	PEELER NO: 3		PEELER NO: 5		PEELER NO: 6	
	SIZE: 4 - 3		SIZE: 4 - 3		SIZE: 4 - 3	
	WOOD: POPLAR		WOOD: POPLAR		WOOD: POPLAR	
	GIRTH	X^2	GIRTH	X^2	GIRTH	X^2
1	26	676	20	400	20	400
2	22	484	21	441	20	400
3	20	400	16	256	24	576
4	24	576	18	324	23	529
5	19	361	25	625	20	400
6	21	441	20	400	20	400
7	23	529	21	441	18	324
8	20	400	21	441	16	256
9	15	225	20	400	19	361
10	17	289	20	400	24	576
11	19	361	22	484	28	784
12	21	441	25	625	26	676
13	23	529	28	784	17	289
14	30	900	31	961	17	289
15	28	784	30	900	24	576

16	26	676	20	400	24	576
17	24	576	16	256	21	441
18	23	529	19	361	25	625
19	25	625	22	484	23	529
20	20	400	15	225	21	441
21	22	484	16	256	17	289
22	15	225	20	400	17	289
23	17	289	21	441	19	361
24	18	324	21	441	28	784
25	21	441	22	484	30	900
26	25	625	21	441	22	484
27	24	576	20	400	26	676
28	20	400	16	256	24	576
29	22	484	18	324	28	784
30	17	289	17	289	20	400
31	15	225	16	256	29	841
32	19	361	20	400	24	576
33	17	289	21	441	23	529
34	18	324	16	256	20	400
35	22	484	17	289	21	441
36	20	400	16	256	18	324
37	18	324	20	400	24	576

38	16	256	22	484	16	256
39	15	225	21	441	19	361
40	16	256	20	400	24	576
AVERAGE	**20.575**	437.075	**20.275**	411.075625	**21.975**	482.900625
Squared Average	423.330625		411.075625		482.900625	
Max	30		31		30	
Min	15		15		16	
S.D	126.8217105		**S.D**	125.0246001	**S.D**	135.5075506
Z1	0.074316929		**Z1**	0.085783118	**Z1**	0.059221792
Z2	-0.04395935		**Z2**	-0.042191697	**Z2**	-0.044093484
A1	0.508		**A1**	0.504	**A1**	0.512
A2	0.484		**A2**	0.484	**A2**	0.484

R1	15
R2	16
R3	14
Subgroups	3
$\bar{X}$	62.825
$\bar{\bar{X}}$	31.4125
$\bar{R}$	15

UCL_x	UCL®
LCL_x	LCL®

S:NO	PEELER NO: 3		PEELER NO: 5		PEELER NO: 6		X Charts		R Charts		Mean
	SIZE: 4 - 3		SIZE: 4 - 3		SIZE: 4 - 3						
	WOOD: POPLAR		WOOD: POPLAR		WOOD: POPLAR						
	GIRTH	X^2	GIRTH	X^2	GIRTH	X^2	UCL_x	LCL_x	UCL_R	LCL_R	$\bar{X}$
1	26	676	20	400	20	400	23.76167	19.171667	23.115	6.885	21.46667
2	22	484	21	441	20	400	23.76167	19.171667	23.115	6.885	21.46667
3	20	400	16	256	24	576	23.76167	19.171667	23.115	6.885	21.46667
4	24	576	18	324	23	529	23.76167	19.171667	23.115	6.885	21.46667
5	19	361	25	625	20	400	23.76167	19.171667	23.115	6.885	21.46667
6	21	441	20	400	20	400	23.76167	19.171667	23.115	6.885	21.46667

7	23	529	21	441	18	324	23.76167	19.1716667	23.115	6.885	21.46667
8	20	400	21	441	16	256	23.76167	19.1716667	23.115	6.885	21.46667
9	15	225	20	400	19	361	23.76167	19.1716667	23.115	6.885	21.46667
10	17	289	20	400	24	576	23.76167	19.1716667	23.115	6.885	21.46667
11	19	361	22	484	28	784	23.76167	19.1716667	23.115	6.885	21.46667
12	21	441	25	625	26	676	23.76167	19.1716667	23.115	6.885	21.46667
13	23	529	28	784	17	289	23.76167	19.1716667	23.115	6.885	21.46667
14	30	900	31	961	17	289	23.76167	19.1716667	23.115	6.885	21.46667
15	28	784	30	900	24	576	23.76167	19.1716667	23.115	6.885	21.46667
16	26	676	20	400	24	576	23.76167	19.1716667	23.115	6.885	21.46667
17	24	576	16	256	21	441	23.76167	19.1716667	23.115	6.885	21.46667
18	23	529	19	361	25	625	23.76167	19.1716667	23.115	6.885	21.46667

19	25	625	22	484	23	529	23.76167	19.171667	23.115	6.885	21.46667
20	20	400	15	225	21	441	23.76167	19.171667	23.115	6.885	21.46667
21	22	484	16	256	17	289	23.76167	19.171667	23.115	6.885	21.46667
22	15	225	20	400	17	289	23.76167	19.171667	23.115	6.885	21.46667
23	17	289	21	441	19	361	23.76167	19.171667	23.115	6.885	21.46667
24	18	324	21	441	28	784	23.76167	19.171667	23.115	6.885	21.46667
25	21	441	22	484	30	900	23.76167	19.171667	23.115	6.885	21.46667

R1	15	**A_2**	0.153	**$\bar{X}$**	64.4	**Subgroups**	3	
R2	16	**D_3**	0.459	**$\bar{\bar{X}}$**	21.46667	**UCL_x** 23.76167	**UCL_R**	23.115
R3	14	**D_4**	1.541	**$\bar{R}$**	15	**LCL_x** 19.17167	**LCL_R**	6.885
AVG	21.56		21.2		21.64			
Max	30		31		30			
Min	15		15		16			

3.4 Calculation of Mean & Range

Upper and lower control limits (UCL_x, LCL_x for X-Bar charts; UCLR, LCLR for R charts) were determined statistically to identify points that fall outside the expected range. Z-scores were computed to assess how many standard deviations a data point is from the mean. Control chart constants (A1, A2) were calculated to aid in the interpretation of the control charts.

3.5 Plotting control chart based on the above data:

Plotting control charts involves graphically representing the collected data and the calculated control limits to visually assess the stability and variability of the manufacturing process. Based on the provided data, control charts can be constructed for X-Bar ($\bar{x}$) and R.

Here's a step-by-step guide to plot the control charts:

3.5.1 X-Bar ($\bar{x}$) Chart:

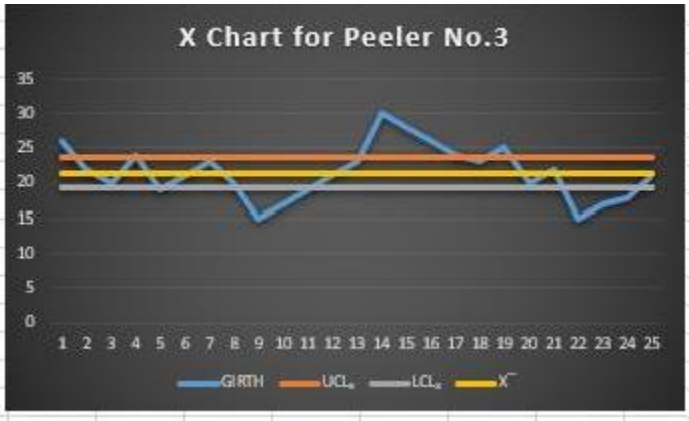
- Calculate the overall mean ($\bar{X}$) and the standard deviation (σ) of the measurements.
- Determine the control limits (UCL_x and LCL_x) for the X-Bar chart based on the calculated mean and standard deviation.
- Plot the individual sample means ($\bar{x}$) on the X-Bar chart.
- Add the UCL_x and LCL_x lines to the chart to visually identify points outside the control limits.

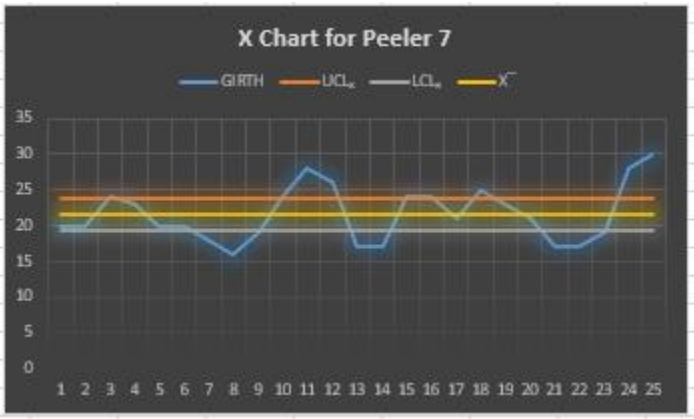
3.5.2 R -Chart:

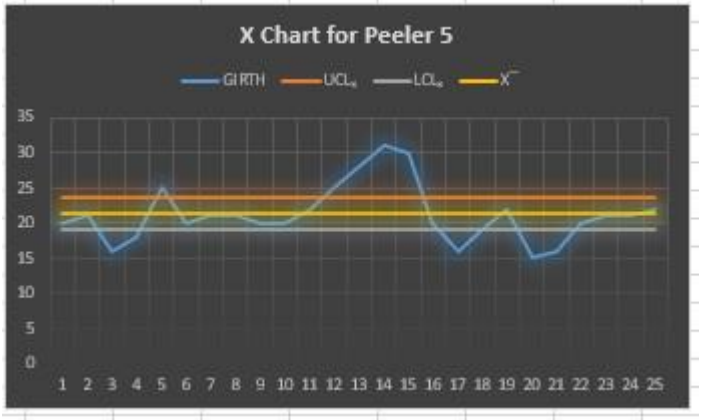
- Calculate the range (R) for each subgroup of measurements.
- Determine the control limits (UCLR and LCLR) for the R chart based on the calculated mean and standard deviation of the ranges.
- Plot the individual subgroup ranges on the R chart.
- Add the UCLR and LCLR lines to the chart to visually identify points outside the control limits.

These steps involve statistical calculations to establish control limits and visually inspecting the charts for any patterns or points exceeding these limits. The goal is to ensure that the manufacturing process is in statistical control and to identify any special causes of variation that may require investigation and corrective action.

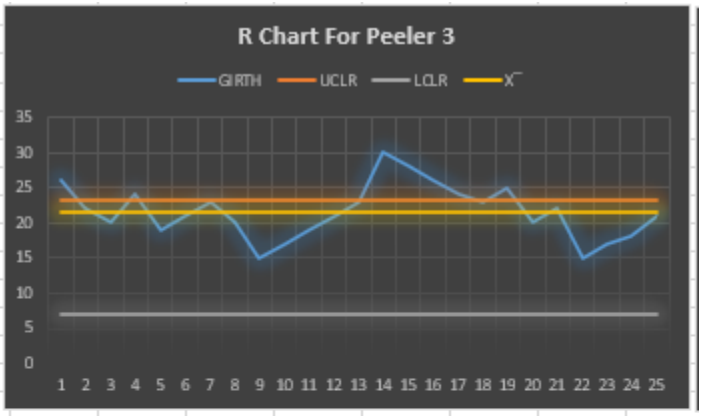
3.1.1 Visual Representation of X charts:

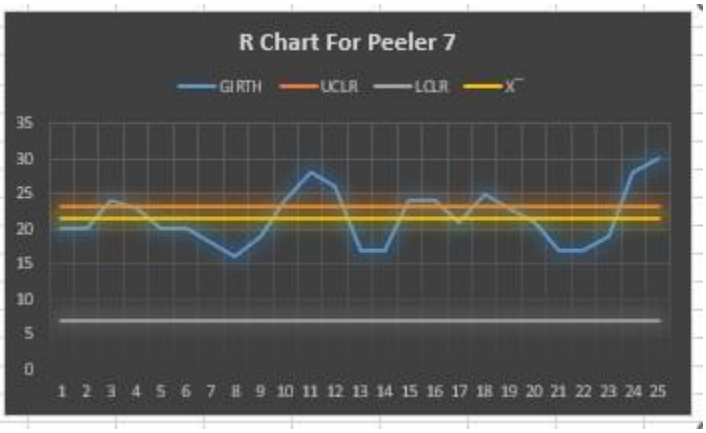


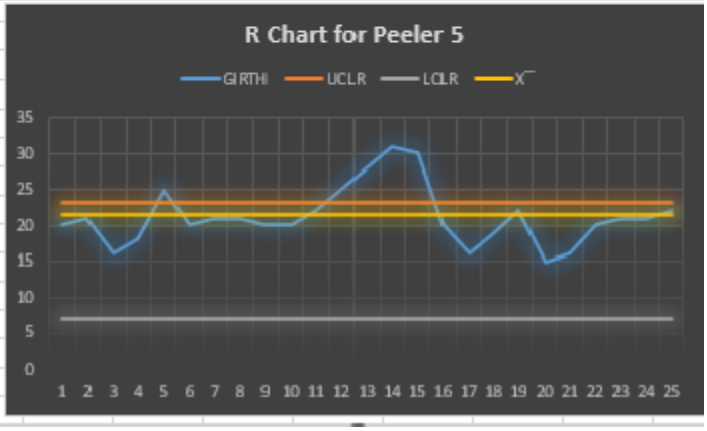




3.1.2 Visual Representation of R chart:







4. Conclusion:

- Control chart technique is a statistical process control method used to detect defects in a production line or process. The technique involves plotting data points on a chart and analyzing them for any patterns or trends that indicate a deviation from the expected performance. Several studies have shown the effectiveness of control charts in detecting defects in various processes, including production lines, distribution cubicles, and wastewater treatment. The use of control charts is often combined with other statistical methods, such as robust regression and principal component analysis, to improve the accuracy of defect detection. Overall, control chart technique is a reliable and widely used method for detecting defects in a process.
- So we have mainly identified that how much of our non-conforming, defective units have been produced. So it is mainly identified with the help of the normal distribution curve in which there is UCL, LCL. So these are mainly made from trial center line in which above and below area are mainly defective parts and this is mainly found and sketched with the help of X and R chart and skewness is also found through this. So these are the main work which we have mainly done in order to compute or defect rate in the production of the ply wood in the ply wood industry.
- The peeler tool has experienced wear and tear, resulting in the production of approximately 37 defective pieces. To address this, it is suggested to replace the old peeler tool with a new one and implement a system
- to monitor and reduce defects by checking the upper and lower control limits for every log. This approach will help in efficiently managing and minimizing the defects in the production process.
- The suggestion to replace the old peeler tool with a new one and implement a monitoring system for controlling defects is essential to ensure the quality and efficiency of the production process. By regularly checking the upper and lower control limits, the defects can be reduced and monitored effectively, leading to improved overall productivity and product quality.

5. Meeting GA7 - Environment and Sustainability

In this project, we have incorporated sustainable manufacturing practices by applying statistical quality control (SQC) methods to evaluate the effectiveness of environmentally friendly processes and their impact on product consistency and reliability. The improvements made during the course of the project contribute significantly to promoting environmental sustainability and reducing the carbon footprint of the manufacturing process.

- **Optimization of Manufacturing Processes:** By utilizing X-Bar and R charts to control the quality of products, we identified areas in the plywood production process that could be optimized. One key improvement was the identification of defective units produced by a worn-out peeler. By replacing the peeler tool with a new one, we are reducing the frequency of defective units, leading to less waste produced in the form of unusable plywood. This directly benefits the environment by reducing material waste, improving the efficiency of resource use, and reducing the energy consumed in the production of defective products.
- **Resource Efficiency:** By implementing the monitoring system based on statistical control charts, we aim to ensure that the manufacturing process is consistent and reliable. This reduces the need for rework and additional resources to correct defects, which directly contributes to minimizing the consumption of raw materials. The result is a more resource-efficient manufacturing process, where each piece of plywood is produced with minimal waste and energy consumption.
- **Sustainable Practices in Tool Management:** Replacing old equipment, such as the peeler tool, is not only a technical improvement for the quality of the product but also an example of incorporating sustainable practices in the maintenance and management of tools and machinery. Regularly replacing worn-out tools ensures the machinery operates efficiently, reduces the need for frequent breakdowns, and leads to fewer defects. This enhances the longevity of machinery and reduces the environmental impact associated with excessive maintenance and the disposal of faulty equipment.
- **Reduced Environmental Impact through Control:** The use of control charts to monitor and maintain stable production processes helps minimize variations that can lead to excess production or unnecessary waste. By focusing on maintaining statistical control over the process, we align with sustainable production goals, ensuring that products meet specifications without excessive overproduction, energy consumption, or waste generation.

6. Meeting PLO7 - Sustainable Development

Sustainable development is at the core of this project, with a focus on applying engineering solutions to improve the long-term performance of manufacturing processes while considering environmental, societal, and economic factors.

- **Addressing Societal and Environmental Contexts:** The improvements proposed in this project are not only beneficial for the company but also have broader implications for society and the environment. By ensuring the quality of plywood products through statistical quality control, we enhance the reliability of

the product and reduce the chances of substandard materials entering the market. High-quality, sustainable products contribute positively to society by providing safer, longer-lasting materials for construction and furniture industries, which in turn supports sustainable urban development.

- **Kaizen Approach for Continuous Improvement:** We employed the Kaizen philosophy of continuous improvement throughout the project. By using statistical methods to monitor quality, we have set a framework for ongoing evaluation and enhancement. This methodology promotes a long-term commitment to sustainability, where the company continuously adapts to new challenges, improves processes, and reduces its environmental footprint over time.
- **Public Health and Safety Considerations:** The improvements made through statistical process control contribute to the safety and health of workers by reducing the risk of defective products being processed. Defective plywood can lead to dangerous products that are unsafe for use in construction or furniture-making. By preventing defects through better monitoring and tool management, the project supports public health and safety by ensuring that only products of the highest quality reach the consumer.
- **Cultural, Societal, and Environmental Considerations:** The project also considers the importance of aligning production practices with cultural and environmental values. By focusing on sustainable manufacturing, we recognize the importance of reducing waste, conserving resources, and ensuring that the production process does not contribute to environmental degradation. These values align with global efforts toward reducing environmental impact, enhancing societal well-being, and contributing to a sustainable future.

7. Summary of Improvements Aligned with GA7 and PLO7

To summarize, the project addresses both GA7 and PLO7 by incorporating principles of sustainability into every phase of the production process. The primary improvements achieved include:

- **Replacement of old equipment** to enhance production quality and reduce waste.
- **Implementation of a monitoring system** based on control charts to ensure process stability and minimize defects.
- **Resource efficiency** in both raw materials and energy consumption by optimizing the production process.
- **Commitment to sustainable development** by continually improving practices and addressing environmental, societal, and public health concerns through engineering solutions.

By focusing on these improvements, this project successfully aligns with the course objectives of promoting sustainability in manufacturing processes, providing engineering solutions that support both economic and

environmental goals.

8. References:

1. Quintero-Arteaga, C., Peñabaena-Niebles, R., Vélez, J. I., & Jubiz-Diaz, M. (2022). Statistical design of an adaptive synthetic $\bar{X}$ control chart for autocorrelated processes. Quality and Reliability Engineering International, 38, 2475-2500.
2. Empirical Comparisons of X-bar Charts when Control Limits are Estimated. J. F. Muñoz-Rosas, E. Álvarez-Verdejo, M. N. Perez-Arostegui, L. Gutierrez-Gutierrez
3. Statistical design of an adaptive synthetic $\bar{X}$ control chart for autocorrelated processes. Carmen Quintero-Arteaga, R. Peñabaena-Niebles, J. I. Vélez, Maria Jubiz-Diaz.